

Haravan: Improving omnichannel sales capabilities in Vietnam with instant Google Cloud scaling power

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ABOUT HARAVAN



About Haravan

Haravan enabled digit growth for V small and medium enterprises, met 100% uptime commitment for clients, handled 5x ecommerce demand spikes, and encouraged

Haravan is a Vietnamese multichannel commerce platform that focuses on making commerce better for everyone, so businesses can focus on building and selling products. Haravan provides a comprehensive software-as-a-service (SaaS) platform for all retail needs, including website creation, logistics, inventory, marketing, and omnichannel sales. Haravan serves more than 50,000 clients in Vietnam, Thailand, and the Philippines. It is a one portfolio company of Seedcom, which recently won \$50

driven expansion Google Cloud.

million in backing from the Ficus investment fund.

Industries: Retail & Consumer Goods

Google Cloud results

- Builds client trust by enabling zero downtime commitment during busiest ecommerce sales periods
- Frees up funds for app development by reducing cloud computing costs by 30%
- Provides scaling power that enables key client to

Absor sales across omnichannel commerce platform

Location: Vietnam

<https://cloud.google.com/kubernetes-engine/>,
Compute Engine



About Cloud Ace

Cloud Ace is a Google Cloud Premier Partner serving the Asia-Pacific with consultation on Google Cloud deployment, architecture, and system optimization. It has over 100 engineers across Japan, Singapore, Vietnam, Thailand, Taiwan, and Indonesia. Cloud Ace won the Google Cloud Application Development Partner of the Year Award in 2018.

Google Cloud (<https://cloud.google.com/>)

Partner serving the Asia-Pacific with

Cloud Storage (<https://cloud.google.com/storage/>)

Preemptible VMs

(<https://cloud.google.com/preemptible-vms/>)

Singapore, Vietnam, Thailand, Taiwan, and

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

achieve 22%
rise in sales

- Inspires new omnichannel ecommerce strategies through AI and data analytics solutions

Compute Engine

(<https://cloud.google.com/compute/>)

Virtual Private Cloud

(<https://cloud.google.com/vpc/>)

Huynh Lam Ho launched his clothing business in 2012, with one shop in Ho Chi Minh City and a website for online sales. But even from the very beginning, he had bigger dreams. He wanted to be running 10+ shops within his second year and an omnichannel sales network that would reach customers through avenues including social media and online marketplaces. His ambitions reflected Vietnam's own impressive growth narrative (<https://www.weforum.org/agenda/2018/09/how-vietnam-became-an-economic-miracle/>)

, riding on entrepreneurial energy to move from poverty to middle-income status within one generation.

Within three months of launching, Huynh had three shops in Vietnam's bustling business capital and a growing online presence. That's when life became a little crazy even for a 24/7 entrepreneur. The team lacked the infrastructure to handle exponential sales growth. "We started to feel the burn of managing inventory, customers, costs, delivery, and suppliers," he recalls.

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*—Hung Le, VP of
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Haravan*

Thanks to their IT backgrounds, Huynh and his partners began building a digital retail platform for internal use, to synchronize all aspects of running the business offline and online. That's when they had a brainwave that would change the future

(<https://www.techinasia.com/haravan-vietnams-shopify-eats-real-shopify-lunch>)

of retail in Vietnam. Why not offer their platform as a software-as-a-service (SaaS) solution to small-and-medium sized retailers facing similar growth pains as them?

They launched Haravan (<https://www.haravan.com/>), quickly attracting merchants to its omnichannel solution that manages every aspect of the client's business. This includes website building, inventory management, logistics, engagement marketing (the use of content to attract customers), and ecommerce over social networks, marketplaces, chat forums, and even livestream.

To achieve the mission, Haravan needed scalable cloud tools to meet its Service Level Agreement (SLA)—the performance guarantee a service makes to clients—of 99.97% platform uptime no matter how much demand spikes from events such as Black Friday and Lunar New Year. It turned to Google Cloud (<https://cloud.google.com/>) and found the tools it needed to build an omnichannel solution that would transform Vietnam's ecommerce landscape.

"We have a guaranteed commitment to enable any volume of sales for clients over all channels, be it social networks, marketplaces, livestream, or website," says Hung Le, VP of Software Engineering at Haravan. "Only Google Cloud, with the flexible autoscaling of Google Kubernetes Engine (/kubernetes-engine) (GKE), gives us certainty to meet our guarantees even in the most massive Black Friday surges."

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Enabling growth ambitions with Google Kubernetes real-time scalability

Haravan began life with on-premises infrastructure, running its microservices in virtual machines (VMs). As the popularity of its service grew across Vietnam, the team soon ran into difficulties managing servers, configuring VMs manually, and scaling to meet the needs of an expanding client base.

In 2016, it won an introduction to Google Cloud for Startups (<https://cloud.google.com/startup>), a program that gives startups access to Google mentors, support networks, and advanced technologies. Through the program, it discovered that by migrating to Google Cloud it could automate many of the mundane DevOps tasks that were eating up development time and save on infrastructure costs that could then be poured into app creation.

In particular, it learned that adoption of a Kubernetes-based architecture on GKE could solve both configuration and scaling challenges that were

hindering expansion. This would be accomplished through automated real-time scalability that both calibrates at instance-level and provides unlimited computing resources.

"Google Cloud for Startups taught us we can do everything in the cloud, with greater flexibility and power and fewer costs, than we could ever dream of on-premises," says Hung. "Google Kubernetes Engine offered exactly the flexible, unlimited scalability we needed for our infrastructure. When our CTO saw the instant scaling power of GKE with his own eyes, he became 100% convinced that Google Cloud was the way to go with our solution."

Discovering the right partner to promote a constantly evolving omnichannel retail solution

After the success of the Google Cloud for Startups Program, Haravan took the evolution of its platform on Google Cloud to a level higher by partnering with Cloud Ace (<https://www.cloud-ace.sg/>). The Google Cloud premier partner advised Haravan on how to build the optimized architecture for its cloud-based SaaS service.

In particular, Cloud Ace guided Haravan to an infrastructure strategy that deploys GKE for the real-time scaling it needs to meet its Service Level Agreement (SLA) obligations, while using preemptible virtual machines (/preemptible-vms) for more routine batch database work. In cloud computing, preemption allows users to run a VM at a much lower cost, with the tradeoff that these

instances can be cut off (or preempted) if computing resources are needed by other clients.

By dividing computing tasks between GKE and preemptible VMs under Cloud Ace guidance, Haravan gained a 30% savings in cloud computing costs, which could then be invested in app development, says Hung.

Deployment of GKE itself, says Hung, proved to be a game changer in enabling Haravan to fulfill its SLAs amid rapid growth both for Haravan and its clients. Hung says GKE enables Haravan to easily absorb a 300% to 500% spike in demand on the busiest days of the year, such as Black Friday, with zero downtime.

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An unlimited future of global growth with Google Cloud AI and ML solutions

In keeping with Vietnam's high-octane business ethos, Haravan is constantly evolving to carve out new paths in the Southeast Asian ecommerce space. It has expanded into Thailand and the Philippines with plans to launch the platform globally.

One key feature of the expansion is Haravan's innovative Harafunnel chatbot. It enables retailers to communicate with customers across the omnichannel network including website, social networks, and livestream events. Harafunnel provides automated responses to customer queries and enables customer acquisition by seeking contact information during livestream chats.

The chatbot currently offers only prefigured responses. But Hung says Google Cloud AI features such as Dialogflow (/dialogflow) and Speech-to-Text (/speech-to-text) may soon turn Harafunnel into a more intelligent solution, solving complex customer issues without deflecting to human customer support.

Meanwhile, Haravan is moving to adopt BigQuery (/bigquery) as its data warehousing solution, to crunch the millions of data points that flow into its system every day. This can provide ML-driven insights on how customers can better target and retain customers through advanced customer relations management (CRM).

"We've already come so far, so quickly, helping Vietnam's retailers to grow with the help of Google Cloud scaling and computing tools," says Hung.

"Who knows how far we'll be able to go worldwide with our omnichannel vision enhanced by Google Cloud AI and data analytics."



